Unit 4 Solutions and Solubility Test - Teacher Version / Answer Key (verA)

Subject: chemistry
Course Code: SCH3U
Unit: Unit 4 Solutions and Solubility Test

Answer Key

1. [solutions_and_concentration | written_response | 3 marks] 0.185 x 750 mL = 139 mL ammonia approximately
2. [solutions_and_concentration | written_response | 3 marks] c = n/V = 0.180/1.50 = 0.120 mol/L
3. [solutions_and_concentration | application | 5 marks] Use C1V1 = C2V2; V1 = (1.50 x 0.5000)/12.0 = 0.0625 L = 62.5 mL. Measure 62.5 mL of stock acid and dilute with water to a final volume of 500.0 mL.
4. [acid_base_neutralization | written_response | 5 marks] H2SO4(aq) + 2KOH(aq) -> K2SO4(aq) + 2H2O(l); n(H2SO4) = 0.00375 mol, n(KOH) = 0.00750 mol, c(KOH) = 0.417 mol/L approximately
5. [solubility_and_solvents | application | 4 marks] Strong responses connect water polarity to dissolving ionic or polar substances as a benefit and explain a drawback such as corrosion, contamination transport, or inability to dissolve non-polar materials well.

Student Paper Reference

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Instructions: Answer all questions. Show full work where required. Write clearly and use appropriate notation.

Written Response

Section Plan: 5 questions | 20 marks

Instructions: Answer all questions. Show all work for calculations, include units, and use appropriate significant digits.

1. A window-cleaning solution is 18.5% (v/v) ammonia. If a bottle contains 750 mL of the solution, what volume of pure ammonia is present? (3 marks)

2. A solution is prepared by dissolving 0.180 mol of sodium carbonate, $Na_2CO_3(s)$, to make 1.50 L of solution. What is the molar concentration of the solution? (3 marks)

3. A stock solution of 12.0 mol/L hydrochloric acid, HCl(aq), is available. What volume of stock solution is needed to prepare 500.0 mL of 1.50 mol/L HCl(aq)? Describe how you would prepare the solution. (5 marks)

4. If 18.0 mL of KOH(aq) is required to completely neutralize 25.0 mL of 0.150 mol/L $H_2SO_4(aq)$, what is the concentration of the base? Write the balanced chemical equation, include states, then solve. (5 marks)

5. Describe one specific example of how water being a good solvent is an advantage and one specific example of how it can be a disadvantage. Explain each example using chemistry ideas. (4 marks)